

Vision-Guided Hearing: An EOG-Based Audio Control System

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Abstract

A three-stage (amplification, band-pass, amplification) circuit was designed to acquire and condition EOG signals and send them to LabVIEW. The LabView VI would then calibrate and determine the thresholds for which direction (left, middle, or right) the user's eyes were pointed. Different audios were played to represent which threshold the signal was in which would then be compared to where the user was actually viewing. EOG is an effective way of determining viewing direction and accurately switching between the three orientations of left, middle, and right. The signal decays to 0 after around 5 seconds making sustained left or right viewing eventually being interpreted as middle viewing. This research provides a proof of concept for selective hearing assistance determined by ocular input.

Introduction

Background Auditory Processing Disorder (APD) is a condition in which the brain cannot effectively filter and interpret sounds despite normal peripheral hearing function. A hallmark symptom is failure of the "cocktail party effect"—the inability to isolate a target voice from competing background noise—leading to hearing fatigue, poor concentration, and reduced quality of life. Around 3–5% of school-aged children are affected by APD, and it frequently co-occurs with ADHD and autism spectrum disorder [1]. These challenges highlight the need for an adaptive, real-time assistive system that can respond to a user's attentional intent in noisy environments.

The eye acts as a biological electric dipole, with the cornea positively charged relative to the retina. As the eye rotates horizontally, this corneo-retinal potential shifts accordingly — producing near-zero differential voltage at center gaze, a positive deflection for rightward gaze, and a negative deflection for leftward gaze, with amplitude linearly proportional to rotation angle within $\pm 30^\circ$ [2]. This feature allows us to track eye movement with relative ease, leading to a technological opportunity when facing APD.

Innovation We propose an electrooculography (EOG)-based instrumentation system that detects horizontal eye gaze direction and uses it to selectively enhance audio from the direction a user is actively attending to. By translating eye movement signals into a real-time gaze classification (Left / Center / Right), the system aims to make busy acoustic environments more accessible for individuals with APD. This project will be a proof-of-concept for vision-based selective hearing that could eventually be incorporated alongside a hearing aid system.

Experimental Setup

Two total skin surface electrodes (placed on the sides of the temples for horizontal reading) read the physical movement of the eyes; the acquired signal is described as 50-3500 μV with frequency content 0-50 Hz [2]. All information will be sent to the DAQ where, after noise filtering and further processing of the dipole signal, eye position can be deduced. With eye position

discovered, the machinery will adjust the volume of the speakers to allow the user to hear the intended noise.

Hardware

Data collection: Electrodes → EOG module → DAQ

Stage 1: AD620 gain $\sim 45\times$, $R_G = 1.1\text{ k}\Omega$

Stage 2: Band-pass filter $\sim (f_{c_highpass} \sim 0.05\text{Hz}) R_1 = 10\text{ k}\Omega$, $C_1 = 350\text{ }\mu\text{F}$;
 $(f_{c_lowpass} \sim 35\text{ Hz}) R_2 = 10\text{ k}\Omega$, $C_2 = 0.45\text{ }\mu\text{F}$.

Stage 3: LM741 gain $\sim 42\times$, $R_i = R_f = 10\text{ k}\Omega$

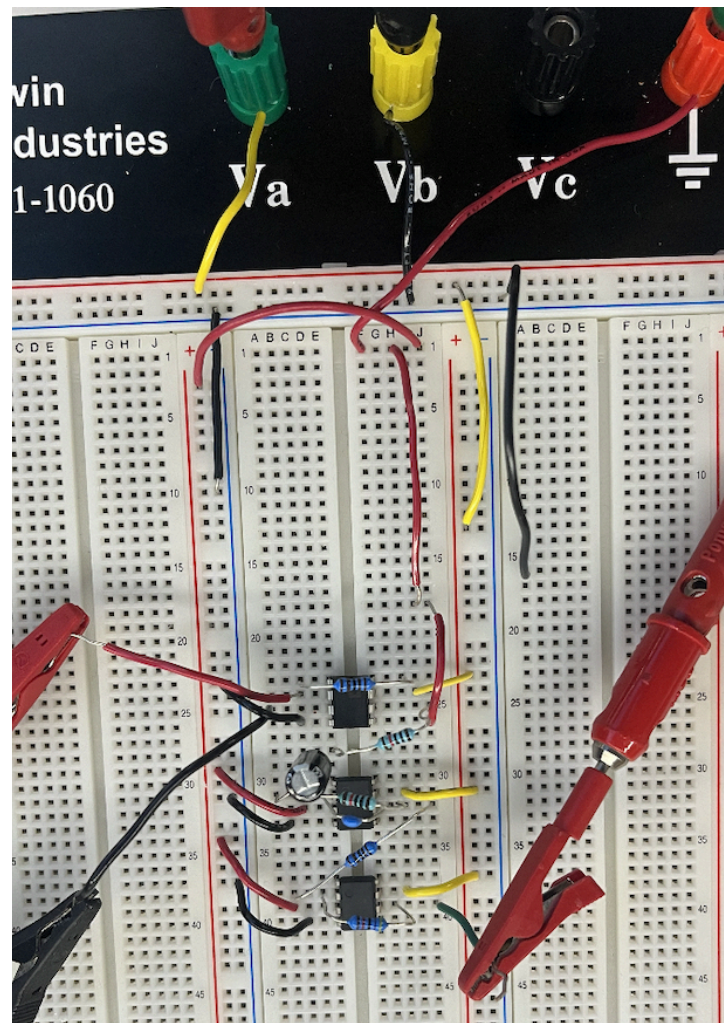


Figure x

Software

As the DAQ—set to collect 100 samples at 1000Hz—sends voltage signals to the computer, LabVIEW first additionally filters any remaining noise before it measures net change in voltage with an integral and boolean logic to determine absolute eye position. This position, expressed with a unique identification number, goes on to modulate an integrated state variable that varies the volume of the 3 sounds associated with the 3 different eye positions. The volume of the

sound associated with the active eye position plays the loudest while the other two are dampened.

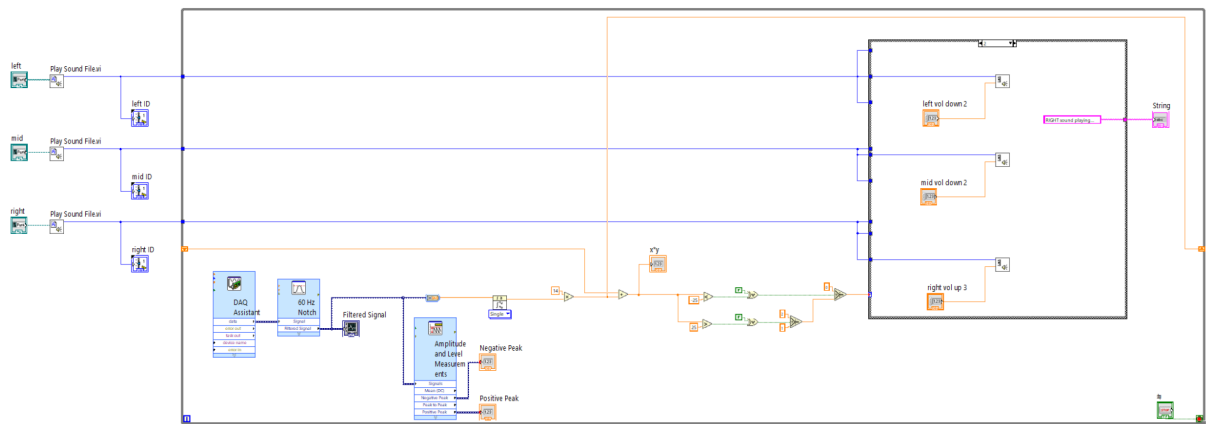


Fig. - x

Testing

A multimeter and oscilloscope was used to measure each selected circuit component as well as validate the expected gain and bandpass capacity. Table X shows overall performance of each component and illustrates the low error system. The bode plot in figure X further validates circuit performance.

Table I
Component Values and Measurement Error

	Rg1 (kO)	Gain 1	Ri (kO)	Ci (μF)	F_high	Rf (kO)	Cf (μF)	F_low	R amp i (kO)	R amp f (KO)	Gain2
Expect ed	1.1	45	10	350	0.047	10	0.45	35.4	1.1	47	42.727 27273
Measur ed	1.097	46.031 9052	10.005	340	0.0467 868839 4	9.983	0.495	32.207 26611	1.095	47.438	43.322 37443
Error (%)	-0.2727 272727	2.2931 22658	0.05	-2.8571 42857	-0.4534 384298	-0.17	10	-9.0190 22293	-0.4545 454545	0.9319 148936	1.3927 91217

Table 1 shows all expected and measured values of each component and resulting error.

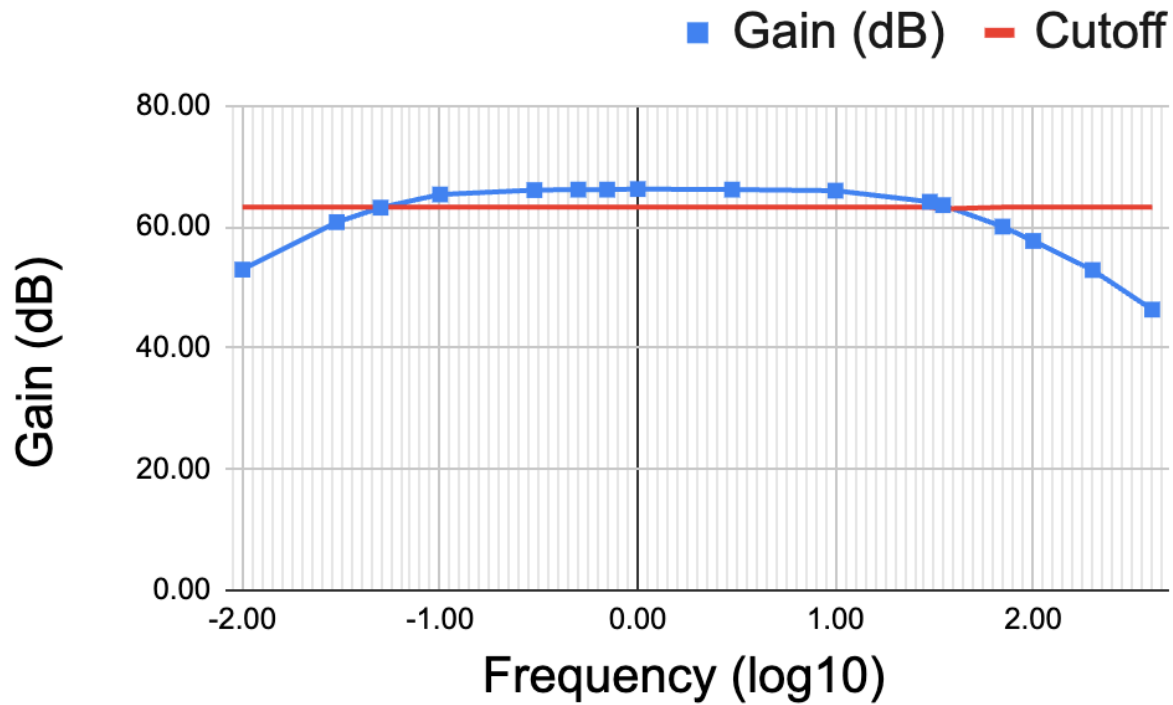


Figure . Bode plot for overall circuit

Results

Table II
Directional Audio Evaluation Results

	Middle to Right	Right to Middle	Middle to Left	Left to Middle	Left to Right	Right to Left
Trial1	Y	Y	Y	Y	Y	Y
Trial2	Y	Y	Y	Y	Y	Y
Trial3	Y	Y	Y	Y	Y	Y

Y denotes that the audio transition in the corresponding direction functions as expected.

The system functioned as expected with all direction transitions correctly triggering the corresponding audio at full volume of 100%, while other directions were attenuated to 10%.

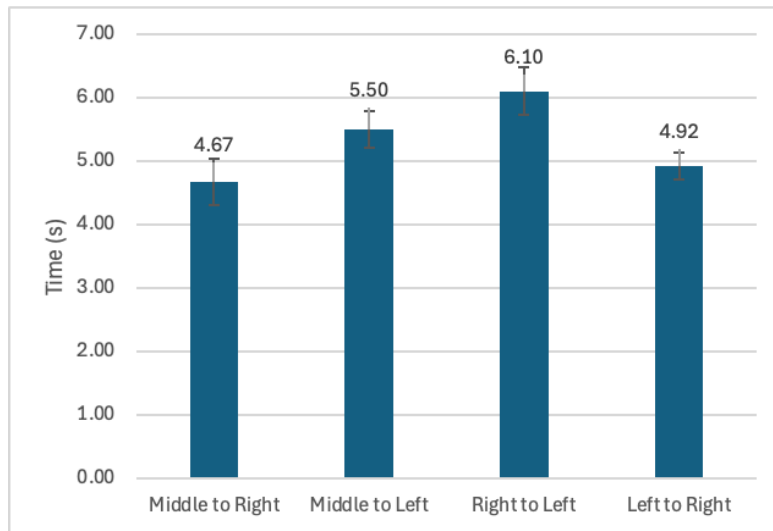


Fig. - Comparison of EOG signal decay time across direction transition. Bars show mean (s) $\pm$ SD; points are individual trials (n = 3). Friedman test indicated a significant overall difference among directions ($\chi^2 = 8.2$, df = 3, p = 0.042).

EOG signals consistently decayed back to baseline after each directional transition, indicating stable system operation (Fig.). The decay time varied across directions (middle to right: 4.67 ± 0.36 s; middle to left: 5.50 ± 0.29 s; right to left: 6.10 ± 0.37 s; left to right: 4.92 ± 0.21 s), with a significant overall difference (p = 0.042).

Discussion and Conclusion

- o What are the key functions of your device?
- o What was the most challenging aspect of the project?
- o If given more time and resources, what improvements could you make?

Key Functions This technology gives users the ability to modulate volume and topic merely with eye movement. In a room with multiple speakers, a patient with APD may not be able to distinguish each individual noise, or may have trouble focusing on just one.

Limitations There were a few limitations our device exhibited, most having to do with the computing limitations of our chosen software. When reading signals into LabView, the initial plan was to run the signals through an integral in order to read the viewing direction instead of the movement of the eyes. However, LabVIEW kept reinitializing the signal which would result in a situation where we would look right, the system would read right, reinitialize, and then read middle despite no changes. We had to adjust our plan by not using an integral and instead reading how the eye movements happened to determine relative switching (for example: from middle to left or from left to right). This in turn introduced the complication of signal decay where all signals would progressively decay towards 0 after a matter of seconds which meant that looking right or left for long enough would eventually be read as middle. This limitation was one

that we were willing to tolerate since it's reasonable to expect the user to turn their head to face what they're looking at, making the interest point the new middle.

Future Improvements The limitations that we faced were due to the logical systems of LabVIEW not storing variables the way that we had intended. We believe that using a different system, such as an Arduino, would allow us to implement our initial logic loop where the signal could be integrated to give the viewing direction. Using the viewing direction would allow us to use the direction threshold method without having to worry about signal decay or reinitialization like in LabVIEW.

- **Application**

- o Discuss the application of your device in the real world

Our device is a proof of concept for using an ocular basis for selective hearing in audio assistance devices. This could in the future be used alongside hearing aid devices to alter directional volume based on the direction that the user glances in to assist in audio isolation. This would be beneficial for patients with APD and the elderly population in reducing the mental strain of daily activity and social interaction.

References (ALL SHOULD BE IN IEEE)